LAB 3: UML CLASS DIAGRAM - MULTI-VENDOR E-COMMERCE PLATFORM

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AIM

To identify the domain entities, attributes, methods, visibility access modifiers, and structural relationships (Generalization, Association, Multiplicity) from the Software Requirements Specification (SRS) of the Multi-Vendor E-Commerce Platform and construct the UML Class Diagram.

THEORY

A Class Diagram is a static structural UML diagram describing the system's schema by displaying classes, attributes, methods, and relationships. Each class contains three compartments: Top (Class Name), Middle (Attributes with visibility: + public, - private), and Bottom (Operations). Key relationships include:

- **Generalization (Inheritance):** A hollow arrow pointing to a superclass representing an *"is-a"* hierarchy (e.g., Customer, Vendor, and Admin inheriting from User).
- **Association & Multiplicity:** Structural links between domain objects indicating instance cardinality (e.g., 1 Customer to 0..* Parent Orders).

IDENTIFIED CLASSES & OPERATIONS

Class	Attributes	Methods
User (Abstract)	- userId: String, - name: String, - email: String, - passwordHash: String, - role: UserRole	+ login(): Boolean, + logout(): Void, + resetPassword(): Boolean
Customer	- customerId: String, - defaultAddressId: String	+ addToCart(productId, qty): Void, + checkout(): Order, + writeReview(): Void
Vendor	- vendorId: String, - businessName: String, - status: VendorStatus, - logoUrl: String	+ createProduct(): Listing, + updateInventory(): Void, + viewEarnings(): Ledger
Administrator	- adminId: String, - permissions: List<String>	+ approveVendor(): Void, + moderateProduct(): Void, + configureCoupon(): Void
Product	- productId: String, - vendorId: String, - name: String, - price: Decimal, - stockQty: Int	+ getDetails(): Product, + updateStock(delta): Boolean
ParentOrder	- orderId: String, - customerId: String, - totalAmount: Decimal, - paymentStatus: Status	+ splitVendorOrders(): List<VendorOrder>, + cancelOrder(): Boolean
VendorOrder	- vendorOrderId: String, - parentOrderId: String, - vendorId: String, - subtotal: Decimal	+ updateStatus(status): Void, + calculateCommission(): Decimal

CLASS DIAGRAM

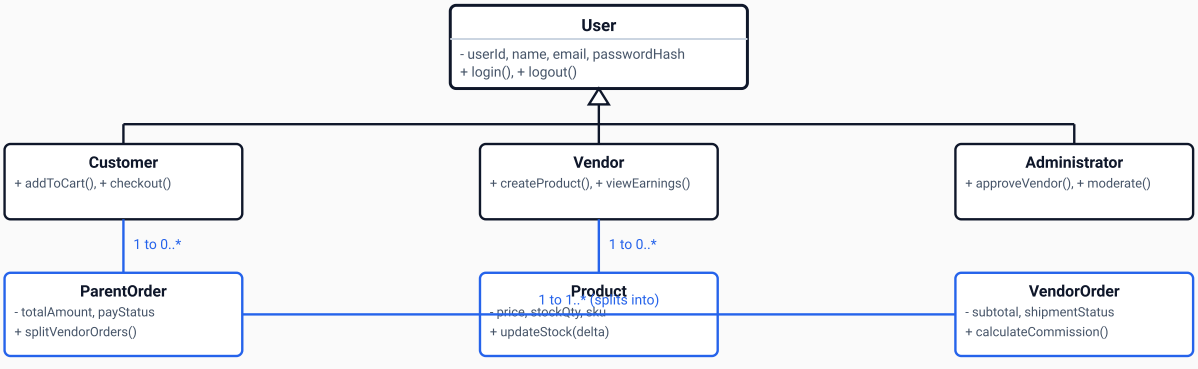


FIGURE 3.1: UML CLASS DIAGRAM OF THE MULTI-VENDOR E-COMMERCE PLATFORM

CONCLUSION

The UML Class Diagram was constructed for the Multi-Vendor E-Commerce Platform. It models role inheritance from a base `User` class and captures associations across Customer checkouts, multi-vendor Order splitting, and Product catalog management.